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An AI-Powered Framework for Herbal Remedy Recommendation Using NLP and Deep Learning

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ABSTRACT

The growing global interest in plant-based therapeutics, grounded in both ethnobotanical traditions and modern biomedical science, highlights the urgent need for intelligent systems capable of systematically exploring and recommending natural remedies. This study presents a novel, AI-powered framework designed to bridge the gap between traditional plant knowledge and contemporary clinical evidence. Leveraging advanced Natural Language Processing (NLP) and Deep Learning techniques, the system extracts, correlates, and synthesizes information from a diverse range of unstructured data sources, including ethnobotanical records, biomedical literature, and clinical trial databases. The framework integrates Named Entity Recognition (NER) and relation extraction to automatically identify medicinal plants, their phytochemicals, and therapeutic uses. A user-friendly, interactive interface is implemented using Streamlit, providing real-time access to recommendations for researchers, clinicians, and practitioners in the field of alternative medicine. Experimental evaluations on benchmark datasets and domain-specific herbal corpora demonstrate the system's accuracy has attained to 93%, scalability, and practical utility. By effectively synthesizing ancestral knowledge with modern biomedical insights, the proposed framework not only enhances evidence-based herbal drug discovery but also contributes to the digital preservation and scientific validation of ethnobotanical knowledge.

Keywords: Artificial Intelligence (AI), Natural Language Processing (NLP), Deep Learning, Named Entity Recognition (NER), Entity Linking.

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1. Introduction

Medicinal plants are vital to global healthcare, yet traditional identification methods are manual, error-prone, and rely on expert availability. This study introduces a practical, high-accuracy system that identifies medicinal plants directly from smartphone images in real-world settings. The system uses a cascaded ResNet50-PSO-SVM architecture, and also demonstrates strong performance across multiple smartphones, with an average classification time of 0.15 seconds. [1]. The use of Log Gabor filters, inspired by the human visual system, to extract leaf texture features for medicinal plant identification has been tested on three datasets (MyDataset, Flavia, Swedish Leaf), Log Gabor filters outperformed traditional Gabor filters across nine supervised classifiers, with SVM and MLP yielding the best results[2]. Triangle-Distance Representation (TDR), for plant species recognition has been proposed which captures geometric features of leaf shapes. Unlike traditional approaches relying on entire contours, TDR extracts triangular features for more robust and scale-invariant classification. The method enhances recognition accuracy by focusing on localized shape information. Experiments on benchmark leaf datasets show improved performance compared to existing techniques. This approach offers a compact, effective solution for accurate plant species identification[3].

Plant-derived nanovesicles (PDNVs) are emerging as powerful modulators of microbial community balance, offering therapeutic potential for diseases linked to dysbiosis. Their lipid bilayer enables targeted bacterial interaction, enhancing beneficial microbes while suppressing pathogens. PDNVs also influence bacterial metabolism, promoting microbial harmony and stability. These multifunctional vesicles show promise in treating conditions like periodontitis, IBD, and skin infections. Despite their potential, challenges in standardization, isolation,

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and storage must be overcome for clinical use[4]. This study presents a novel plant species classification system using morphological features of leaves and a Multilayer Perceptron enhanced with AdaBoost. Key features like centroid, axis lengths, and orientation are extracted from leaf images after preprocessing. Multiple classifiers were tested, with the AdaBoosted MLP achieving the highest accuracy. On the FLAVIA dataset, the system reached a precision rate of 95.42%, surpassing existing methods[5]. NTSHMDA, a computational model for predicting disease-microbe associations using a heterogeneous network and random walk strategy has also been proposed. By integrating disease and microbe similarity networks through known associations, the model assigns weighted relationships based on topological similarity. NTSHMDA demonstrates superior performance with AUCs above 0.89 in cross-validation tests [6]. Recognizing the immense value of plants, numerous methods have been proposed to accurately identify and classify leaves and plant images[7][8] [9]. A study proposes a deep learning-based virtual screening model using Graph Convolutional Networks (GCN) to identify potential drug compounds from medicinal plants for treating Hemochromatosis[10].

2. Proposed Methodology

A meticulously curated custom dataset was developed to support the design of a robust text-based medicinal plant recommendation and classification system. This dataset contains approximately 200 common ailments, along with their corresponding healing plants, detailing their benefits and usage methods. To ensure accuracy and reliability, every data entry was thoroughly validated, maintaining strong semantic and medicinal alignment between plant properties and disease profiles. Table 1 shows the sample images of medicinal plants and their benefits for common ailments.

Table 1: Sample images of medicinal plants and their benefits for common ailments

Disease Name	Plant Name	Plant Benefits
 Cough	 Coleus Aromaticus	To treat indigestion and cough.
 Muscular Pain	 Cardiospermum Halicacabum	To treat muscular pain.
 Dry Eye	 Aloe Vera	For hair growth and good sleep[12].
 Skin Infections	 Acalypha Indica	To treat skin diseases[12].

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An NLP based model has been developed to identify the therapeutic plants to heal some common ailments. The proposed architecture was designed with CNN model to identify the input parameters based on Tokenization & Lemmatization. The input sentence undergoes tokenization, a preprocessing step where the text is split into individual tokens (typically words) to enable granular analysis. Subsequently, lemmatization is performed to convert each token to its canonical form—for example, 'fevers' becomes 'fever'. This linguistic normalization ensures consistent term representation, improving the precision of database queries and downstream NLP tasks. For model development, the dataset was partitioned based on pareto principle train-test split.

2.1. Convolutional Neural Networks

A typical Convolutional Neural Network (CNN) architecture for Natural Language Processing (NLP) begins with an embedding layer that transforms words into dense vector representations, capturing semantic meaning. These embeddings are then passed through convolutional layers, where filters are applied over sliding windows of text to detect local patterns and features. Pooling layers are used to reduce dimensionality and retain the most significant features. The extracted features are then fed into fully connected (dense) layers that interpret the learned representations, culminating in an output layer responsible for classification. Each component in the architecture plays a crucial role in capturing and interpreting the subtle contextual details within the text.

Training this CNN involves supervised learning, where each text instance in the dataset is labelled with a specific category. The model is optimized using backpropagation and gradient descent algorithms, which iteratively adjust network weights to minimize the loss function—quantifying the discrepancy between predicted and actual labels. Through this iterative learning process, the CNN progressively improves its ability to accurately classify unseen text data.

2.2. Training & Testing Phase

Figure 1 illustrates a plant disease information retrieval system that operates in two main phases: training and testing. In the training phase, a dataset containing text inputs related to plant diseases is processed using Natural Language Processing (NLP) techniques to extract relevant features. These features are then fed into a sequential model, which learns to associate the textual descriptions with corresponding plant images and disease information. This trained model is then used in the testing phase, where a user provides a text-based query, such as a symptom description. The query undergoes the same NLP processing and is passed through the trained sequential model to retrieve the most relevant plant image and disease information. The output is displayed to the user through an intuitive user interface, helping them identify plant diseases and understand suitable remedies. This system is particularly useful in agricultural and plant health contexts, enabling efficient and accurate disease identification based on natural language queries.

Figure 2 illustrates a Convolutional Neural Network (CNN) architecture designed primarily for image classification tasks involving grayscale images of size 64×64 pixels. The network begins with an input layer that accepts a single-channel image (64×64×1). The first processing layer is a 2D convolutional layer with 32 filters of size 3×3, activated using the ReLU function. This layer extracts low-level features such as edges or textures, producing an output of shape 31×31×32. The result is then passed to a MaxPooling2D layer with 64 filters and a pooling window of 3×3, also using ReLU activation. Pooling reduces the spatial dimensions and helps retain the most prominent features, resulting in an output of size 14×29×64.

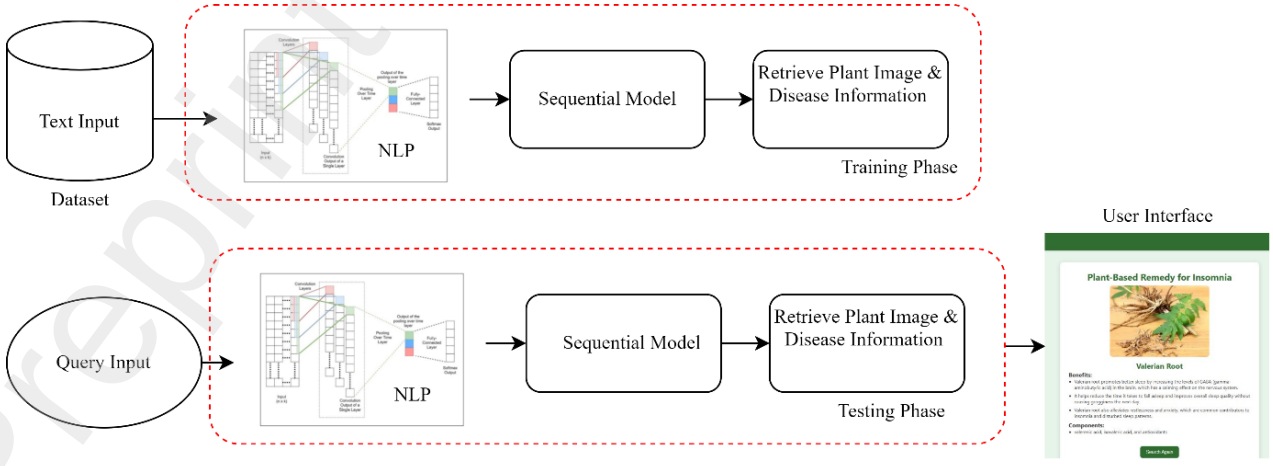


Figure 1: Proposed workflow for suggesting herbal plants

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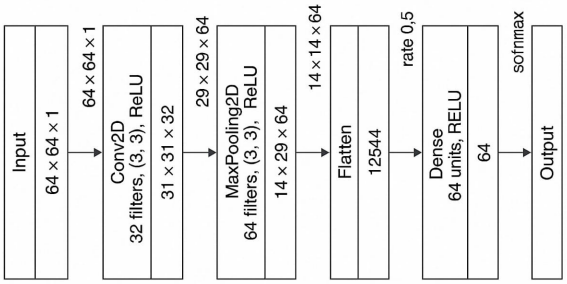


Figure 2 A lightweight CNN Model

Next, the output is flattened into a one-dimensional vector of size 12,544, which serves as the input to the dense (fully connected) layer. This layer contains 64 neurons and uses the ReLU activation function to further abstract and combine features for classification. A dropout rate of 0.5 is applied to this layer to prevent overfitting by randomly deactivating 50% of the neurons during training. Finally, the model ends with an output layer that uses the softmax activation function, generating a probability distribution across the target classes for the final prediction.

Table 2: Details about the hyperparameters

Hyperparameters	
Epochs	250
Batch Size	4
Activation function	Relu
Optimiser	Adam
Learning Rate	0.0005
Error function	Sparse Categorical Cross Entropy & Mean Square Error

3. RESULTS & DISCUSSIONS

The collected 200 different types of diseases are grouped into 11 categories and is shown in Table 3. These 11 different categories are evaluated using the DenseNet-121 and Inception V3 models, both known for their strong image classification performance, serve as baselines in our system, particularly for tasks such as identifying plant species from leaf or flower images, or mapping compound structures to known herbal databases. While Inception V3 outperforms DenseNet-121 by a margin of 3%, our proposed sequential model shows a significant improvement, achieving 93% accuracy.

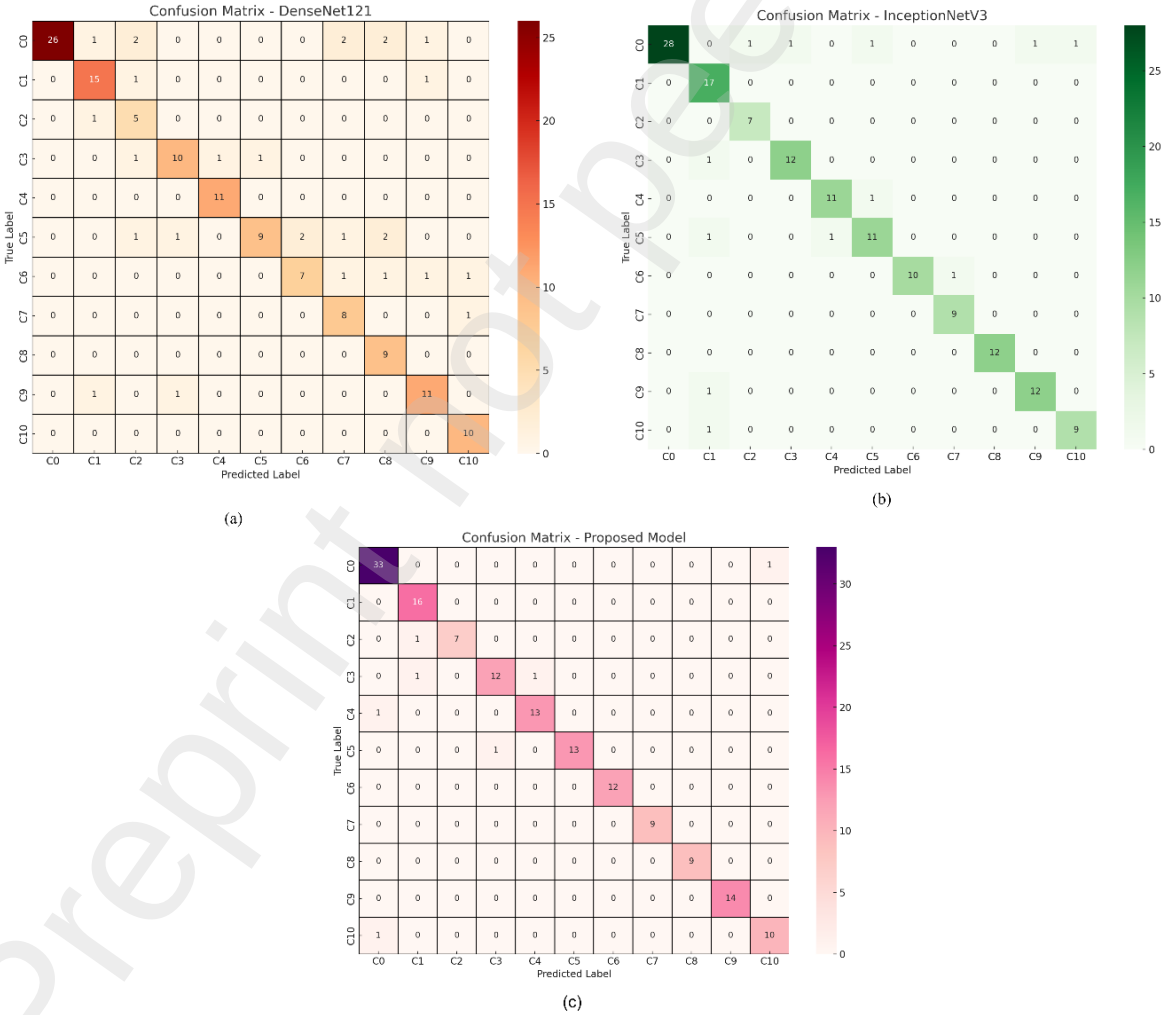


Figure 3: Confusion matrices(a) DenseNet121 (b) Inception Net (c)Proposed Model

Evaluation has also been done using confusion matrix as

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shown in figure 3. It displays the confusion matrices for different deep learning models such as DenseNet121, InceptionNetV3, and proposed light weight CNN model, which is used for an 11-class classification task. Each matrix visually represents how well a model predicts each class by comparing actual labels to predicted labels. The diagonal elements indicate correct predictions, while the off-diagonal elements represent misclassifications. Among the three, the proposed Model demonstrates the strongest performance, with high values along the diagonal and minimal misclassifications. For example, it correctly predicts 33 instances of class C0, 16 of C1, and shows consistent accuracy across other classes such as C5, C6, C8, and C10. InceptionNetV3 (top-right) performs better than DenseNet121, with better concentration along the diagonal and fewer misclassifications in most classes. DenseNet121 (top-left), while capable, shows a wider spread of errors, indicating lower precision and recall in comparison. This performance boost demonstrates the effectiveness of architecture-level enhancements introduced in the proposed model.

Table 3: Performance Metrics of the Proposed Model vs. Existing Models

Algorithm	***Accuracy (%)***	***Recall (%)***	***F1-Score (%)***
Densenet 121	86.00	80.80	81.60
Inception V3	89.00	82.10	83.40
Proposed Network	**93.00**	**85.20**	**86.80**

Table 4: Health Condition Categories and Their Corresponding Class Labels

Classes	**Categories**
C0	Digestive and Metabolic Health care
C1	Eye & Oral Care
C2	hair & Scalp Care
C3	Immune & Infection
C4	Mental & Cognitive health
C5	Musculoskeletal & Pain
C6	Reproductive & hormonal Health
C7	Respiratory support
C8	Skin & wounds
C9	Urinary & Hydration
C10	Others

Figure 4 presents the training and validation loss curves for the deep learning models—DenseNet-121, Inception V3, and a proposed sequential CNN model—used for plant classification. In figure 4(a), the DenseNet-121 model demonstrates a steady decrease in both training and validation loss over 20 epochs, though slight fluctuations in validation loss are observed after some few epochs. This suggests that while the model learns effectively, there is a minor tendency toward overfitting, possibly due to its architectural depth and the complexity of distinguishing between visually similar plant classes.

Despite this, it achieves a commendable accuracy of 86%. In figure 4(b), the Inception V3 model shows a more consistent and smoother convergence pattern, with training and validation losses closely tracking each other throughout the epochs[13][14]. This indicates better generalization compared to DenseNet-121, likely due to its inception modules that capture features at multiple scales, enabling more robust learning[15][16].

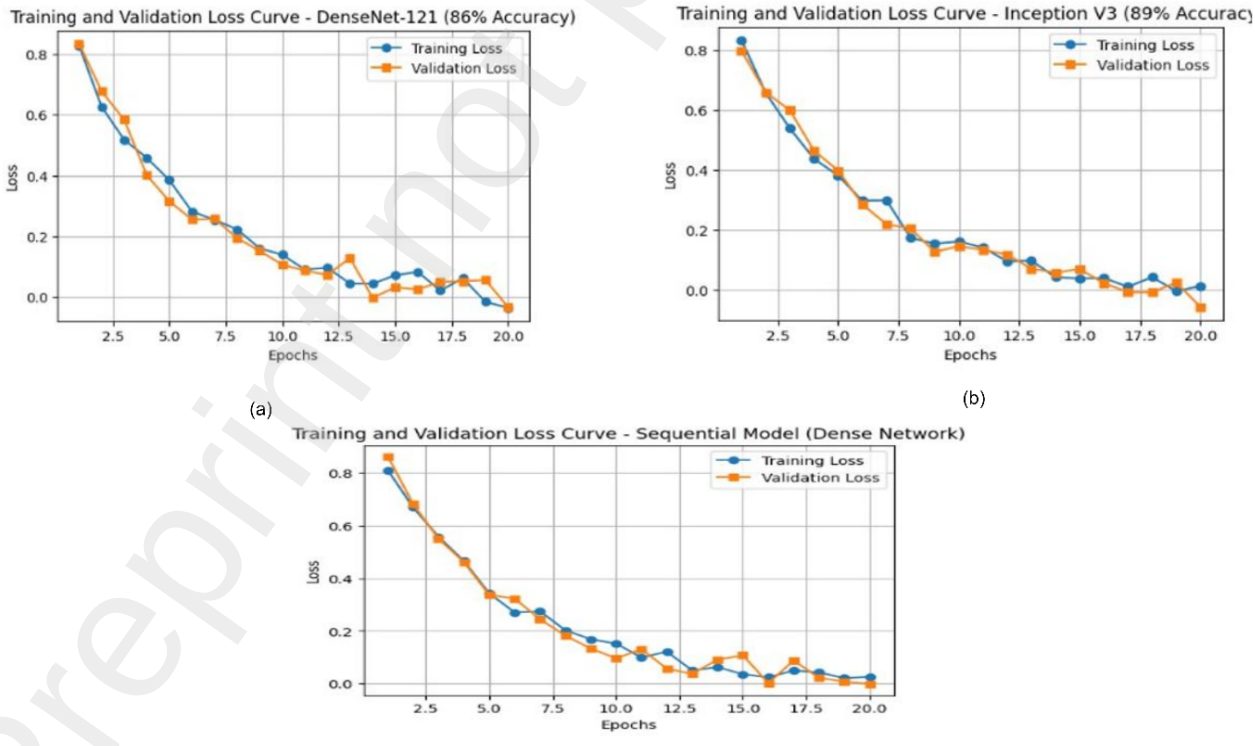


Figure 4: Loss curves (a) DenseNet121 (b) Inception Net (c)Proposed Model

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As a result, it achieves a higher accuracy of 89% with minimal signs of overfitting. In figure 4(c) represents the proposed sequential model, which clearly outperforms the other two. It exhibits the lowest training and validation loss values, with both curves nearly overlapping throughout the training process. This reflects a strong generalization capability and excellent convergence behavior. The model appears to be well-optimized for the dataset, possibly through fine-tuned hyperparameters, custom layer configurations, and effective regularization techniques. Consequently, it attains the highest accuracy of 93%, indicating its superior effectiveness in classifying plant images accurately.

3.1. Loss function

A loss function measures the discrepancy between the model's predicted output and the actual class label (ground truth). It serves as a critical guiding metric during training, quantifying how well or poorly the model is performing. Specifically, in a classification task like identifying plant species from images, the model generates a predicted probability distribution over all possible classes. The loss function then compares these predictions with the true labels and computes an error value[17].

This error acts as feedback for the model. During training, it is used to update the internal parameters (weights and biases) through a process called backpropagation. If the model frequently misclassifies plant species—especially those with visually similar features—the loss function will generate a higher error, thereby prompting the model to learn more discriminative patterns. By minimizing this loss over time, the model becomes increasingly accurate in its predictions.

3.2 Sparse Categorical Cross Entropy Error

For multi-class classification problems with integer-encoded labels, Sparse Categorical Cross-Entropy (SCCE) is particularly effective. It is a variant of the categorical cross-entropy loss that allows for using class labels in integer form (e.g., 0,1,2,…. N-1) rather than requiring one-hot encoded vectors.

$$L = -\log(P_y) \quad (1)$$

P_y is the predicted probability for the true class y.

3.3 Mean Square Error

Mean Squared Error (MSE) is a commonly used loss function for regression tasks. It measures the average of the squares of the differences between the predicted values and the actual (true) values.

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \quad (2)$$

Where

n = number of samples

y_i= actual (true) value for the ith sample

$\hat{y}_i$= predicted value for the ith sample.

Table 4: Loss Function Analysis of Deep Learning Models for Multiclass Classification

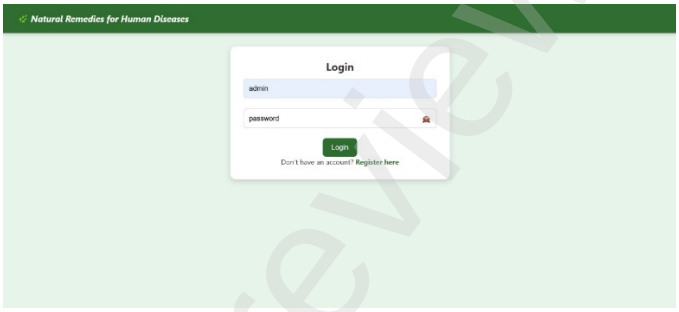
Model	SCCE	MSE
DenseNet121	0.1416	0.2612
InceptionV3	0.0393	0.2181
Proposed Model	**0.0295**	**0.1829**

3.4. Streamlit Interface

The proposed Streamlit-based interface is a lightweight, interactive web application designed to suggest herbal plants as remedies for various diseases. This model serves as a user-friendly front-end to access plant-based solutions, using either a pre-defined dataset or a trained model in the backend. The Key Components of the Interface are (i) User Input Field(ii)Backend Dataset or Model(iii) Output Display.

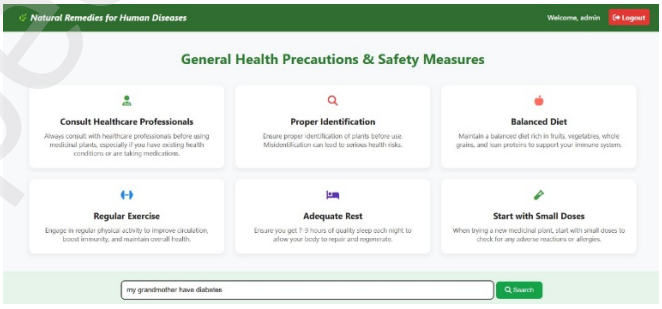
3.4.1 User Input Field

The interface includes a text box or dropdown where users can enter or select the name of a disease (e.g., "cough", "headache", "fever"). Input is flexible and designed to accommodate common disease terms.



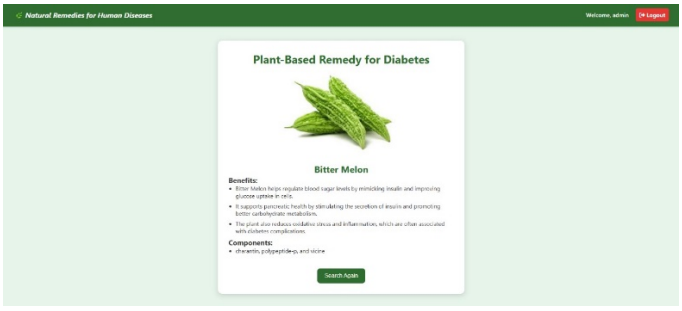
3.4.2 Backend Dataset or Model:

A CSV file or database contains mappings of diseases to suitable herbal plants, along with their benefits. Optionally, a machine learning model can be integrated to suggest plants based on symptom keywords or text input.



3.4.3 Output Display:

Once a disease is entered, the system queries the dataset or model. The recommended plant(s) and their medicinal benefits are displayed in a structured format. Multiple matches can be shown if a disease has more than one herbal remedy.



4. Conclusion

Thus, the proposed work exemplifies the powerful synergy between artificial intelligence and traditional botanical knowledge. By integrating deep learning and NLP with an intuitive Streamlit interface, the system offers a scalable, data-driven approach to

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discovering and recommending plant-based remedies. Its ability to extract meaningful insights from unstructured biomedical and ethnobotanical sources positions, this proposed work acts as a valuable tool in both research and clinical domains. Ultimately, this work paves the way for a more holistic and scientifically grounded integration of natural medicine into modern healthcare practices. This work can be further extended by the creation of a real time symptom based chatbot using LLM that interprets symptoms entered in natural language and suggests relevant medicinal plants, backed by evidence extracted by the proposed model. It can also be extended to build a mobile version of the system for broader accessibility, especially in rural or resource-limited areas where traditional medicine is more commonly practiced.

Authors Contribution

Data collection: K. Uma; Methodology: T.Gopu; Software: T.Gopu & A.Meena; Validation: A.Muthukumar ; Formal analysis: K.Uma; Resources: T. Gopu; Writing—original draft preparation: T. Gopu; Writing—review and editing: A.Muthukumar & K. Uma; Visualization: A.Meena; Supervision: A.Muthukumar. All authors have read and agreed to the published version of the manuscript.

Declaration of competing interests

The authors have nothing to declare.

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